EUNJIN CHOI

Curriculum Vitae (Last update: May 12, 2026)

Ph.D Candidate @ Music and Audio Computing Lab, KAIST

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RESEARCH INTERESTS

Music Generation, Music Representation, Music Information Retrieval

As a researcher, my work focuses on enabling machines to learn and understand the compositional rules of human-created music. I have been working on controllable symbolic music generation guided by musical constraints, as well as symbolic music representation learning with self-supervised approaches. As a developer, I'm interested in adapting MIR technologies into real-world applications such as game. Ultimately, my goal is to empower creative musicking by developing technologies that foster and amplify human creativity and expression.

EDUCATION

Ph.D. Student in Culture Technology Mar. 2022 – Present
Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea
Advisor: Juhan Nam

M.S. in Culture Technology Mar. 2020 – Feb. 2022
Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea
Thesis: YM2413-MDB: A Collection of Multi-instrumental Symbolic Music and 1980s FM Music Generation for Video Game
Advisor: Juhan Nam

B.E. in Convergence Mar. 2015 – Feb. 2020
Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, Korea
Graduated magna cum laude

JOURNALS

Mel2Word: A Text-based Melody Representation for Symbolic Music Analysis
Saebyul Park, **Eunjin Choi**, Jeounghoon Kim, Juhan Nam
Music and Science, 2024

CONFERENCES

D3PIA: A Discrete Denoising Diffusion Model for Piano Accompaniment Generation
Eunjin Choi, Hounsu Kim, Hayeon Bang, Taegyun Kwon, Juhan Nam
2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2026

Cutting Kim: Playful Transgression Through VR Voice Interaction in Public Exhibition Contexts
Joonhyung Bae, **Eunjin Choi**, Juhan Nam
SIGGRAPH Asia 2024 Art Papers, ACM, 2025

AImoclips: A Benchmark for Evaluating Emotion Conveyance in Text-to-Music Generation
Gyehun Go, Satbyul Han, Ahyeon Choi, **Eunjin Choi**, Juhan Nam, Jeong Mi Park
Proceedings of the 3rd Workshop on Human-Centric Music Information Research, 2025

On the De-duplication of the Lakh MIDI Dataset
Eunjin Choi, Hyerin Kim, Jiwoo Ryu, Juhan Nam, Dasaem Jeong
Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR), 2025

PianoBind: A Multi-modal Joint Embedding Model for Pop-piano Music
Hayeon Bang, **Eunjin Choi**, Seungheon Doh, Juhan Nam
Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR), 2025

PIAST: A Multi-modal Piano Dataset with Audio, Symbolic and Text
Hayeon Bang, **Eunjin Choi**, Megan Finch, Seungheon Doh, Seolhee Lee, Gyeong-Hoon Lee, Juhan Nam
Proceedings of the 3nd Workshop on NLP for Music and Spoken Audio (NLP4MuSA), 2024

Teaching Chorale Generation Model to Avoid Parallel Motions
Eunjin Choi, Hyerin Kim, Juhan Nam, Dasaem Jeong
Proceedings of the 16th International Symposium on Computer Music Multidisciplinary Research (CMMR), 2023

Bridging Audio and Symbolic Piano Data through a Web-Based Annotation Interface
Seolhee Lee, **Eunjin Choi**, Joonhyung Bae, Hyerin Kim, Eita Nakamura, Dasaem Jeong, Juhan Nam
Late Breaking Demo in the 24th International Society for Music Information Retrieval Conference (ISMIR), 2023

YM2413-MDB: A Multi-Instrumental FM Video Game Music Dataset with Emotion Annotations
Eunjin Choi, Yoonjin Chung, Seolhee Lee, Jong Ik Jeon, Taegyun Kwon, Juhan Nam
Proceedings of the 23rd International Society for Music Information Retrieval Conference (ISMIR), 2022

Harmonionz, Rescue The Planet: A Voice Visualizing Game that Match Pitch with Color
Hajun Kim*, **Eunjin Choi***, Byeoli Choi* (*Equal Contribution)
2021 Annual Symposium on Computer-Human Interaction in Play, 2021

*All authors contributed equally to this work.

DOMESTIC CONFERENCES

Cutting Kim: VR Voice Interaction Game
Joonhyung Bae, **Eunjin Choi**, Juhan Nam
Proceedings of the HCI Korea 2023, 2023

EXPERIENCES

Neutune, South Korea Jun. 2021 – Sep. 2021
Research Intern, Symbolic Music Generation

INFINYX, South Korea Jul. 2019 – Aug. 2019
Research Intern in AI Development Team, Medical Image Segmentation (gallbladder)

Visual Computing Lab, DGIST, South Korea Jan. 2019 – Feb. 2019
Research Intern, Super-Resolution and Denoising

Music and Audio Research Group, Seoul National University, South Korea Jul. 2018 – Nov. 2018
Research Intern, Music Cognition Study and Musical Syntax Experiment

Stem Cell Institute, University of Cambridge, United Kingdom Jan. 2018 – Feb. 2018
Research Intern, Studying Lung Stem Cell Tumor Development

UC Berkeley, United States Jun. 2016 – Aug. 2016
Summer Session, Coursework: Introduction to Music Theory (Grade A)

TEACHING EXPERIENCES

Creating Games with Unity Apr. 2023 - Dec. 2023
Main Instructor, KAIST SW Education Center

Computational Analysis of Music Mar. 2022 - Dec. 2022
Teaching Assistant, Daejeon Science High School R&E

Investigation of Mozart Dice Music Generation System Mar. 2021 - Dec. 2021
Teaching Assistant, Daejeon Science High School R&E

Game Studies (GCT544) Sep. 2020 - Dec. 2020
Teaching Assistant, KAIST Graduate School of Culture Technology

EXHIBITIONS & WORKSHOPS

Exhibition, ACT Festival 2023: Future Legends, Asia Culture Center, Gwangju, Korea — Nov. 2023

Workshop, 2022 Daejeon Art and Science Biennale: The Future City Associated Program - Game and Art: Future City of Fantasy, Daejeon Museum of Art, Daejeon, Korea — Oct. 2022

INVITED TALKS

"Introduction to Music Information Retrieval", DGIST — Mar. 2023

"Introduction to Music Transcription and Audio and Symbolic-level Music Analysis", DGIST — May. 2024

"The Making Process of Harmonionz, Rescue the Planet", KAIST — Sep. 2021

AWARDS AND HONORS

Awardee, Producer DK AI Composition Challenge — 2025

3rd Place, KAIST Anthem Challenge — 2025

NC Scholarship, NCSoft — 2022

1st Prize, KAIST x NCSoft Game Design Course — 2020

Finalist, Unity Game Development Challenge (G-STAR 2018) — 2018

DGIST Undergraduate Research Award — 2018

1st Prize, Poster Session, DGIST Undergraduate Research Program — 2017

Hye-seul Leadership Award, DGIST — 2017

Academic Excellence Award, DGIST — 2016

SERVICES

Organizing Committee, Local Organization Chair, ISMIR — 2025

Accountant, Korean Society for Music Perception and Cognition (KSMPC) — 2022 - 2024

Exhibition Assistant, Game and Art: Auguries of Fantasy explores 'Fantasy Reality' — Jun. 2021 - Sep. 2021

Reviewer, ISMIR, ICASSP, IEEE TASLP, KSMI

LANGUAGES, SKILLS, AND INTERESTS

English (fluent), Korean (native)

Python(PyTorch), C, C++, C#, Matlab, HTML/CSS, JavaScript, Unity, VR (Oculus Quest 2)

Piano, violin, acoustic guitar, music composition

REFERENCES

References available upon request